Equation of a Straight Line

Exam code: 9MA0

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Help me understand

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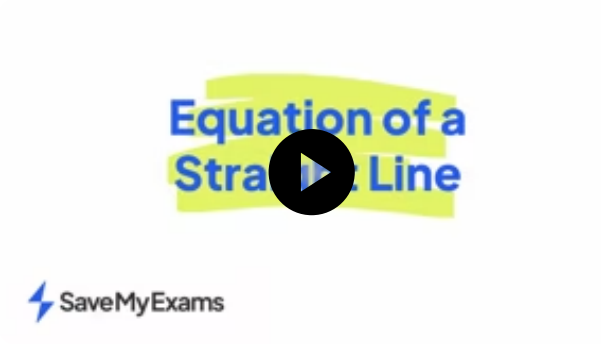

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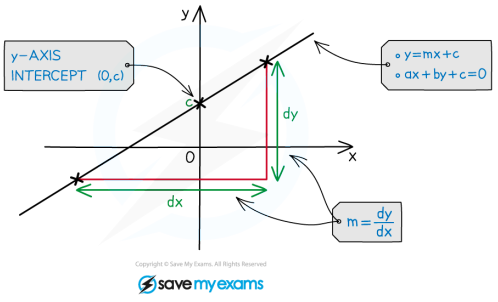
Equation of a straight line

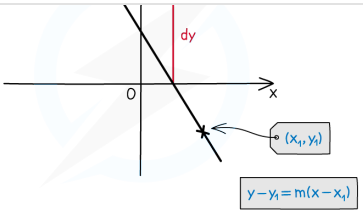
What is the equation of a straight line?

- **y = mx + c** is the equation for any straight line
- **m** is **gradient** given by "difference in **y**" ÷ "difference in **x**" or **dy/dx**
- **c** is the **y**-axis **intercept**
- Alternative form is **ax + by + c = 0** where **a**, **b** and **c** are integers



How do I find the equation of a straight line?

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



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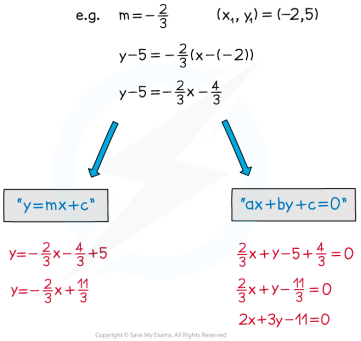
- **Two** features of a straight line are needed
  - gradient, **m**
  - a point the line passes through, (x_1, y_1)
- The equation can then be found using $y - y_1 = m(x - x_1)$
- This can be rearranged into either $y = mx + c$ or $ax + by + c = 0$



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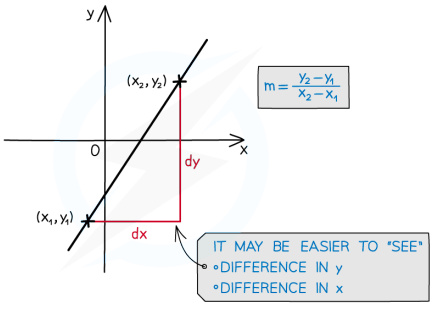
How do I find the gradient of a straight line?

- There are lots of ways to find the gradient of a line
- Using t**wo points** on a line to find the change in y divided by change in x

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$



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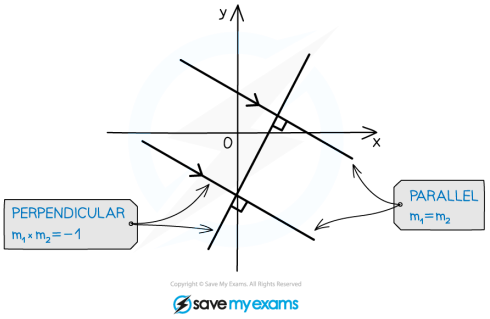
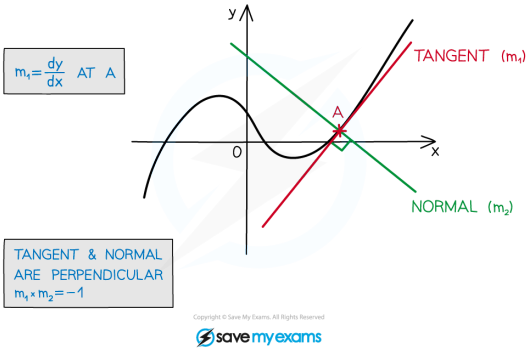


- Using **Tangents** and **Normals** - **Differentiation** (see Gradients, Tangents & Normals)



- Other ways
  - **Collinear** lines are the same straight line so gradients are equal
  - Angle facts and **circle theorems**
    eg. a **radius** and **tangent** are **perpendicular**

Examiner Tips and Tricks

- Working with straight lines can involve lots of algebra, but **sketching** a diagram will always help.
- Use a **sketch** to check if answers seem about right.

Worked Example

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